Week 5 Lab Session: Exploratory Data Analysis (EDA) & Dashboard Creation

Duration: 2 hours
Objective: To introduce students to data preprocessing and exploratory data analysis using Python, focusing on the Titanic dataset.

🛠️ Prerequisites

- Basic knowledge of Python programming.
- Access to the internet and a web browser.
- Optional: Python installed locally with libraries such as pandas, numpy, matplotlib, seaborn.

📝 Lab Activities

1. Introduction to EDA and the Titanic Dataset (10 minutes)

- Brief overview of Exploratory Data Analysis (EDA) and its importance in data science.
- Introduction to the Titanic dataset: understanding the features and the objective of predicting survival.

2. Guided Walkthrough: Titanic EDA Tutorial (30 minutes)

- Navigate to the Titanic EDA tutorial on Kaggle:
  - Titanic EDA Tutorial
- Follow along with the tutorial, covering:
  - Data loading and initial exploration.
  - Handling missing values.
  - Feature engineering.
  - Data visualization techniques.
- Encourage students to run the code snippets and observe the outputs.

3. Hands-On Practice: W3Schools Python ML Basics (30 minutes)

- Explore the W3Schools Python Machine Learning tutorials:
  - W3Schools Python ML Getting Started
- Topics to cover:

    - Mean, Median, Mode.
    - Standard Deviation.
    - Percentiles.
    - Data Distribution.
- Students practice these concepts using sample datasets provided in the tutorials.